

PATENT APPLICATION

THE PATENTS ACT, 1970 (39 of 1970) & THE PATENTS RULES, 2003

— COMBINED FORMAT —

AI-POWERED ETHICAL DECISION SUPPORT SYSTEM FOR INFORMATION TECHNOLOGY PROFESSIONALS

Application Type:	Ordinary Application
Under Act:	The Patents Act, 1970 (39 of 1970) — Section 7
Field (IPC):	G06F 21/00 — Security Arrangements for IT; G06N 20/00 — Machine Learning
Category:	Natural Person / Educational Institution
Reference:	CSE332 — Industry Ethics and Legal Issues

FORM 1 — APPLICATION FOR GRANT OF PATENT

THE PATENTS ACT, 1970 (39 of 1970) & THE PATENTS RULES, 2003

[See Section 7, 54 and 135 and Sub-rule (1) of Rule 20]

1. Applicant's Ref No.:	CSE332/AIEDS/2025-26
2. Type of Application:	Ordinary (☒) Convention (☐) PCT-NP (☐) Divisional (☐)

3A. APPLICANT(S) DETAILS

Field	Details
Name (Full)	[Your Full Name Here]
Gender	[Male / Female / Other]
Nationality	Indian
Country of Residence	India
Age	[Your Age] years
Address	[House No., Street, City, State, PIN Code]
Email	[Your Email ID]
Contact No.	[Your Mobile Number]

3B. Category:	Natural Person (☒) Educational Institution (☒) Startup (☐)
4. Inventor(s) same as Applicant?	Yes (☒) No (☐)
5. Title of Invention:	AI-Powered Ethical Decision Support System for IT Professionals
6. Patent Agent:	Not Applicable (Self-filing)
7. Address for Service:	[Your Institution/College Address, City, PIN, Email]
8. Convention Country:	Not Applicable
9. PCT Application:	Not Applicable
10. Divisional Application:	Not Applicable

12. DECLARATION

I/We, the above-named applicant(s), hereby declare that: I am/We are in possession of the above-mentioned invention. The complete specification relating to the invention is filed with this application. I am/We are the true and first inventor(s) of this invention. There is no lawful ground of objection to the grant of the Patent to me/us. The invention as disclosed in the specification is novel, involves an inventive step, and is capable of industrial application.

Date: _____

Signature: _____

Place: _____

Name: _____

To, The Controller of Patents, The Patent Office, at New Delhi / Mumbai / Chennai / Kolkata

FORM 2 — COMPLETE SPECIFICATION

THE PATENTS ACT, 1970 (39 of 1970) & THE PATENTS RULES, 2003
[See Section 10 and Rule 13]

1. Title of Invention:	AI-Powered Ethical Decision Support System for IT Professionals
2. Applicant(s):	[Your Name] — Indian National — [Your Address]

3. PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

The present invention relates to the field of information technology ethics and information security. More specifically, it pertains to an artificial intelligence-based real-time decision support system that monitors, evaluates, and guides the ethical and legal behavior of information technology professionals during the execution of their professional duties, including but not limited to code development, database access, cloud system configuration, and data management activities.

BACKGROUND OF THE INVENTION / PRIOR ART

The rapid growth of information technology has created significant challenges in ensuring ethical and legally compliant behavior among IT professionals. Existing compliance tools, such as Data Loss Prevention (DLP) systems, firewalls, and post-incident audit systems, operate reactively — they detect violations only after the harmful action has already been executed. This reactive approach results in data breaches, regulatory penalties, reputational damage, and legal liabilities that could have been prevented.

Current systems suffer from the following critical limitations:

1. No real-time ethical guidance at the point of action for IT professionals.
2. Existing tools block actions without explaining the legal or ethical reason.
3. No system provides a contextual risk score based on role, jurisdiction, and data sensitivity.
4. No natural language explanation is provided to guide the user toward a compliant alternative.
5. Audit trails are generated retroactively rather than proactively at the decision point.

Research by Poszler & Lange (ScienceDirect, 2024) and ACM (2023) identifies real-time ethical decision support in IT professional workflows as a significant and unaddressed research gap. The present invention addresses this gap by providing a proactive, explainable, and workflow-integrated ethical decision support system.

OBJECTS OF THE INVENTION

- i. To provide a real-time AI-powered system that prevents unethical and illegal IT actions before they are executed.
- ii. To assign a contextual ethical risk score to each IT professional action based on role, data sensitivity, applicable law, and jurisdiction.

- iii. To provide natural language explanations of ethical/legal violations and suggest compliant alternatives.
- iv. To generate automatic audit trails at the point of decision for legal defensibility.
- v. To integrate seamlessly into existing IT professional workflows such as code editors, cloud consoles, and database access panels.
- vi. To support compliance with applicable laws including GDPR, the Indian IT Act 2000, and organizational data policies.

SUMMARY OF THE INVENTION

The present invention provides an AI-Powered Ethical Decision Support System (AI-EDSS) for IT professionals. The system comprises a Behavioral Pattern Analysis Engine, a Contextual Ethical Scoring Module, a Legal Rule Engine, a Natural Language Explanation Generator, and an Audit Trail Generator. The system operates in real time, intercepting IT professional actions before execution, evaluating them against ethical principles and applicable legal frameworks, computing a risk score, and providing actionable guidance to the user.

Unlike existing compliance tools that audit actions retrospectively, the present invention intervenes proactively at the moment of action — preventing violations before damage occurs, explaining the reason for flagging in plain language, and suggesting a legally compliant alternative. This novel combination of real-time ethical inference, legal rule engine, and professional workflow integration constitutes the inventive step of the present invention.

BRIEF DESCRIPTION OF DRAWINGS

Fig. 1 — System Architecture Overview of the AI-Powered Ethical Decision Support System.

Fig. 2 — Process Flow Diagram showing real-time interception and ethical evaluation of IT professional actions.

FIG. 1 — SYSTEM ARCHITECTURE DIAGRAM

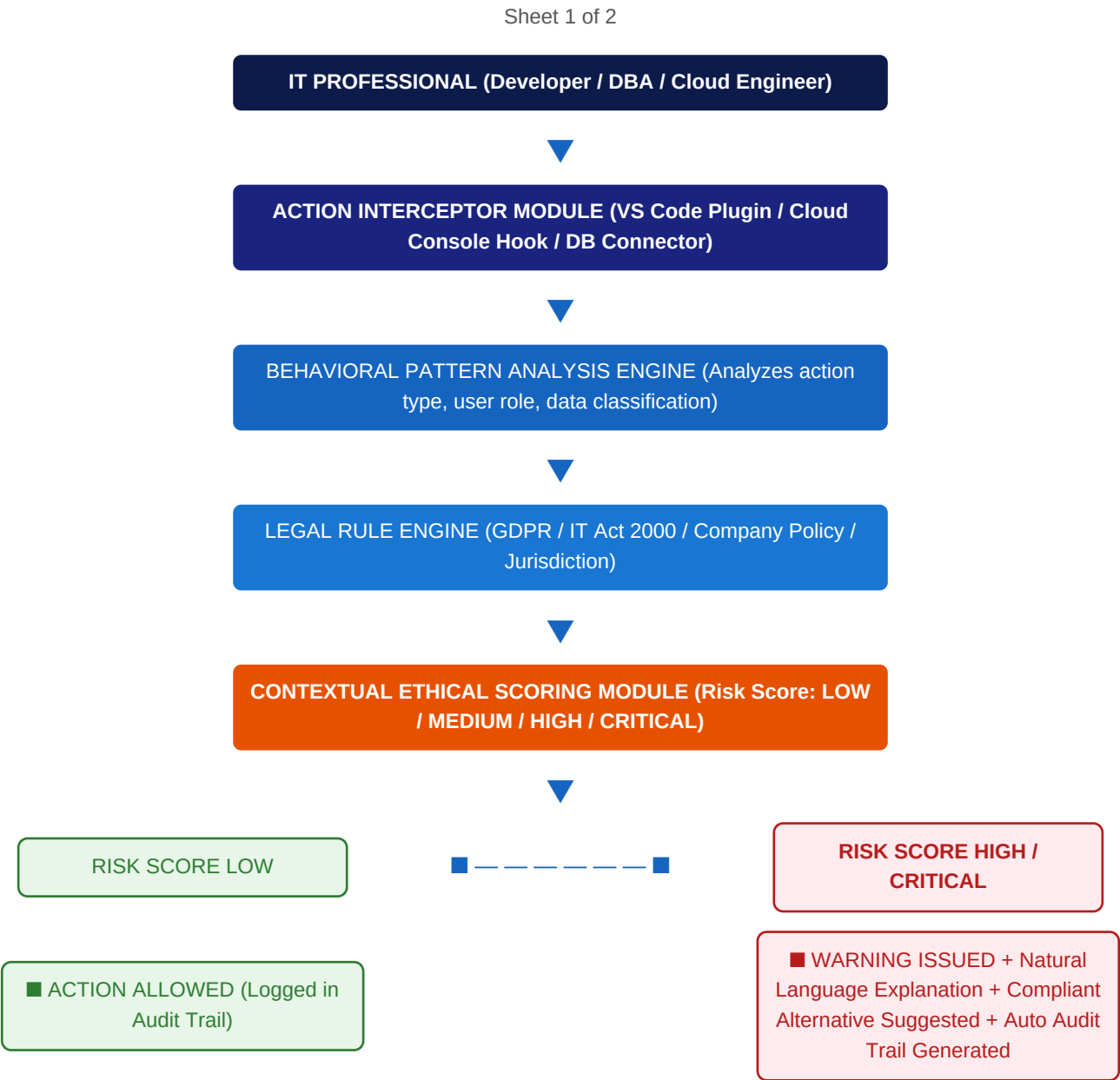


FIG. 1 — System Architecture of AI-Powered Ethical Decision Support System

FIG. 2 — PROCESS FLOW DIAGRAM

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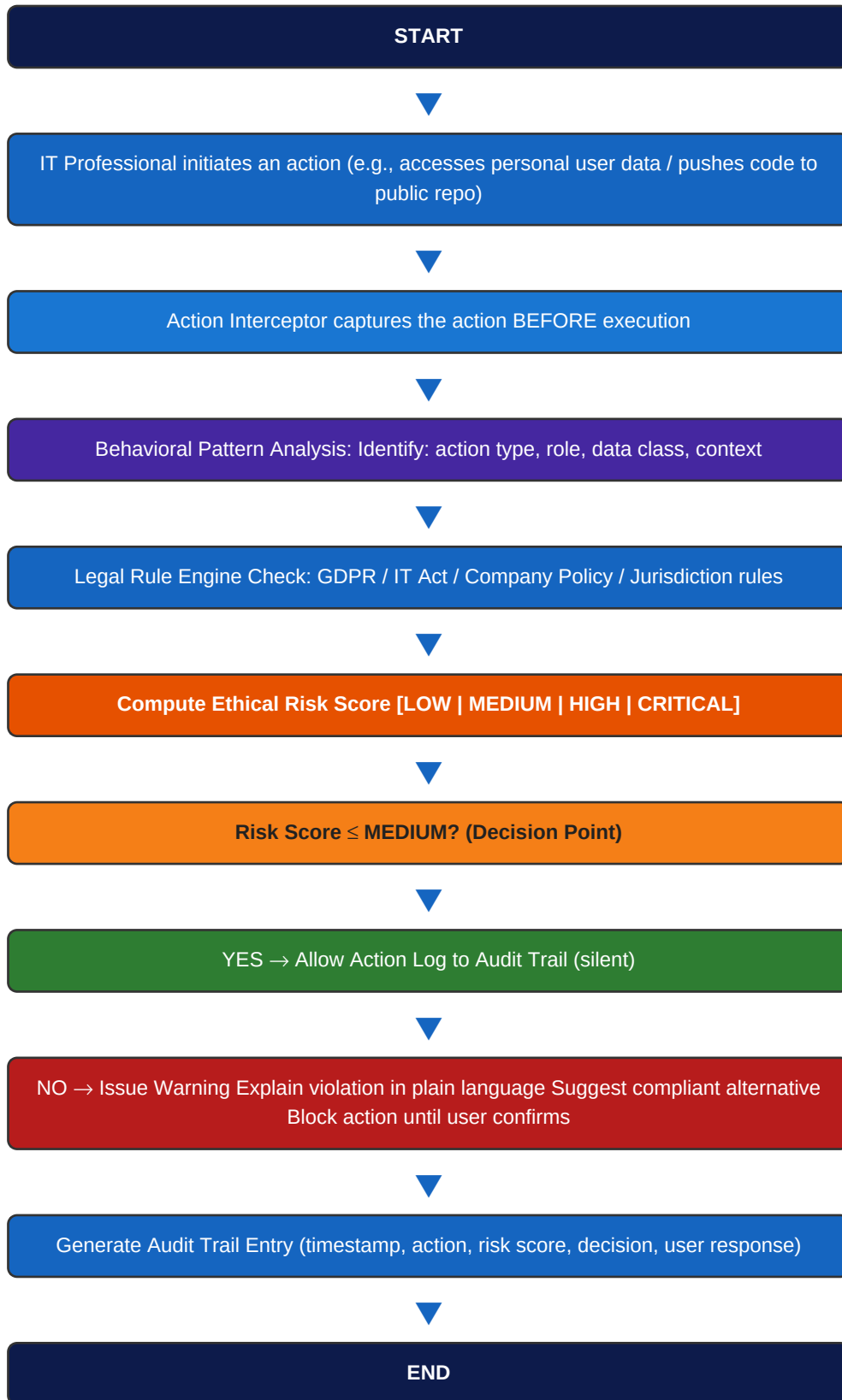


FIG. 2 — Process Flow Diagram of AI-Powered Ethical Decision Support System

DETAILED DESCRIPTION OF THE INVENTION

100. System Overview

The AI-Powered Ethical Decision Support System (AI-EDSS), hereinafter referred to as 'the System', is a software-based invention that operates in real time within the professional workflow of an IT professional. The System comprises five core modules: (i) Action Interceptor Module [101], (ii) Behavioral Pattern Analysis Engine [102], (iii) Legal Rule Engine [103], (iv) Contextual Ethical Scoring Module [104], and (v) Natural Language Explanation and Audit Module [105].

101. Action Interceptor Module

The Action Interceptor Module [101] is implemented as a plugin or extension integrated into professional IT tools including but not limited to code editors (e.g., Visual Studio Code), cloud management consoles (e.g., AWS Console, Azure Portal), and database access panels. The module captures every action initiated by the IT professional before execution and passes it to the Behavioral Pattern Analysis Engine [102] for evaluation.

102. Behavioral Pattern Analysis Engine

The Behavioral Pattern Analysis Engine [102] employs machine learning algorithms to analyze the captured action in context. The engine considers: (a) the type of action being performed, (b) the role and access level of the IT professional, (c) the classification and sensitivity of the data involved, and (d) historical behavioral patterns of the user. The output is a structured action-context vector passed to the Legal Rule Engine [103].

103. Legal Rule Engine

The Legal Rule Engine [103] contains a rule base comprising applicable legal frameworks including but not limited to the General Data Protection Regulation (GDPR), the Information Technology Act 2000 (India), the IT (Amendment) Act 2008, and configurable organizational data policies. The engine matches the action-context vector against the rule base and generates a policy-violation assessment.

104. Contextual Ethical Scoring Module

The Contextual Ethical Scoring Module [104] computes a numerical Ethical Risk Score (ERS) on a scale of 0 to 100, classified into four bands: LOW (0-25), MEDIUM (26-50), HIGH (51-75), and CRITICAL (76-100). The ERS is computed as a weighted function of: (a) severity of potential legal violation, (b) data sensitivity level, (c) professional role authorization level, and (d) jurisdictional context.

105. Natural Language Explanation and Audit Module

When the ERS exceeds the MEDIUM threshold, the Natural Language Explanation and Audit Module [105] performs the following operations: (a) generates a plain-language explanation of the detected violation, (b) cites the specific law or policy being violated, (c) suggests one or more compliant alternatives, and (d) creates a timestamped audit trail entry recording the action, ERS, explanation issued, and the user's subsequent decision. When the ERS is within the LOW-MEDIUM band, the action is permitted and silently logged.

EXAMPLE OF OPERATION: A software developer (Role: Junior Developer, Access Level: Read-Only) attempts to execute a database query that retrieves the personal identification numbers of all registered users. The Action Interceptor Module [101] captures this query before execution. The Behavioral Pattern Analysis Engine [102] identifies this as a bulk personal data access action by a user whose role does not include bulk data export privileges. The Legal Rule Engine [103] matches this against GDPR Article 5 (data minimisation principle) and the organization's data access policy. The Contextual Ethical Scoring Module [104] computes an ERS of 82 (CRITICAL band). The Natural Language Explanation and Audit Module [105] issues the following warning: 'This action may violate GDPR Article 5 — Data Minimisation. Your role does not permit bulk export of personal identification data. Suggested alternative: Use the anonymised_users view instead.' The action is blocked pending user acknowledgment, and an audit trail entry is generated.

CLAIMS

(Each claim is a complete single sentence as required under the Patents Act 1970)

WE CLAIM:

1. [Independent Claim]

A computer-implemented AI-powered ethical decision support system for IT professionals, comprising: an Action Interceptor Module configured to capture IT professional actions before execution; a Behavioral Pattern Analysis Engine configured to analyze the captured action in context of user role, data sensitivity, and historical behavior; a Legal Rule Engine containing rules corresponding to applicable data protection and information technology laws; a Contextual Ethical Scoring Module configured to compute a numerical Ethical Risk Score; and a Natural Language Explanation and Audit Module configured to issue plain-language warnings, suggest compliant alternatives, and generate audit trails when the Ethical Risk Score exceeds a predefined threshold.

2. [Dependent on Claim 1]

The system as claimed in Claim 1, wherein the Legal Rule Engine contains rules corresponding to the General Data Protection Regulation (GDPR), the Information Technology Act 2000 of India, and configurable organizational data policies.

3. [Dependent on Claim 1]

The system as claimed in Claim 1, wherein the Contextual Ethical Scoring Module computes the Ethical Risk Score as a weighted function of severity of potential legal violation, data sensitivity level, professional role authorization level, and jurisdictional context.

4. [Dependent on Claim 1]

The system as claimed in Claim 1, wherein the Action Interceptor Module is implemented as a plugin integrated into code editors, cloud management consoles, or database access panels.

5. [Dependent on Claim 1]

The system as claimed in Claim 1, wherein the Natural Language Explanation and Audit Module generates a timestamped audit trail entry recording the action details, Ethical Risk Score, explanation text, and user's subsequent decision, for purposes of legal defensibility.

6. [Dependent on Claim 3]

The system as claimed in Claim 3, wherein the Ethical Risk Score is classified into four bands: LOW (0-25), MEDIUM (26-50), HIGH (51-75), and CRITICAL (76-100), and actions in the HIGH and CRITICAL bands are blocked pending user acknowledgment.

7. [Method Claim]

A computer-implemented method for providing real-time ethical decision support to IT professionals, comprising the steps of: intercepting an IT professional action before execution; analyzing the action using behavioral pattern analysis considering user role and data sensitivity; checking the action against a legal rule engine containing applicable data protection laws; computing a contextual Ethical Risk Score; and when the score exceeds a threshold, issuing a natural language explanation of the violation, suggesting a compliant alternative, and generating an audit trail entry.

ABSTRACT OF THE INVENTION

(Not more than 150 words — on a separate page as required)

The present invention relates to an AI-Powered Ethical Decision Support System (AI-EDSS) for Information Technology professionals. The system comprises an Action Interceptor Module, a Behavioral Pattern Analysis Engine, a Legal Rule Engine, a Contextual Ethical Scoring Module, and a Natural Language Explanation and Audit Module. The system intercepts IT professional actions — including code commits, database access, and cloud configurations — before execution, evaluates them in real time against applicable legal frameworks (GDPR, IT Act 2000) and organizational policies, computes a contextual Ethical Risk Score, and issues plain-language warnings with compliant alternatives when violations are detected. Unlike existing post-incident compliance tools, the present invention prevents ethical and legal violations proactively, while generating automatic audit trails for legal defensibility. The invention is applicable in IT companies, banks, healthcare systems, and cloud platforms.

[Word Count: ~130 words — within the 150-word limit as prescribed]

Date:

Signature of
Applicant:

Place:

Name (in full):

"An invention that does not just detect unethical actions — but prevents them, explains them, and teaches better decisions."